Code No: 138BJ **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year II Semester Examinations, June - 2024
DATA ANALYTICS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What are the drawbacks of randomized block design?	[2]
b)	What is the difference between transaction data and reference data?	[3]
c)	What is the need of analytics?	[2]
d)	Give examples for MNAR and MCAR category.	[3]
e)	What is meant by residual in linear regression?	[2]
f)	List the three amin purposes for using linear regression.	[3]
g)	What is unsupervised learning?	[2]
h)	What is a regression tree?	[3]
i)	Give examples for complex relations.	[2]
j)	Illustrate the usage of Chernoff faces.	[3]

PART – B

(50 Marks)

2. Describe the general approach of designing data architecture at three levels of specifications. [10]

OR

3. Differentiate between noise and outlier. Describe techniques to handle them. [10]

4. Does repeated imputation improve the quality of estimation? Justify your answer with suitable examples. [10]

OR

5. Explain the difference between reporting and analytics. Enumerate various applications of analytics. [10]

6. With appropriate data set demonstrate ordinary least square technique for linear regression. [10]

OR

7. Compare and contrast linear regression and logistic regression. [10]

8. Illustrate the working of CHAID algorithm for constructing decision tree using suitable data. [10]

OR

9. What are the various components of time-series data? Explain them with examples.[10]

10. Describe the importance of data visualization and illustrate geometric projection visualization. [10]

OR

11. Which domains need data visualization applications for complex data? Explain suitable applications and domains. [10]

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